Security Posture Assessment of Brazilian Municipal Web Portals

A Comprehensive Analysis of the 26 State Capitals and the Federal District

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Abstract -
This study presents a comprehensive security analysis of 27 Brazilian municipal portals, encompassing all 26 state capitals and the Federal District. Through systematic evaluation conducted on November 3, 2025, of SSL/TLS configurations, security headers implementation, and service exposure patterns, we identified significant security gaps in fundamental protection mechanisms. Our methodology employed automated scanning tools and manual validation to assess multiple security dimensions across all Brazilian geographic regions. Key findings reveal that only 22.2% of municipal websites implement HTTP Strict Transport Security (HSTS), while 44.4% operate with suboptimal SSL/TLS configurations (Grade B or below). The research establishes baseline security metrics for Brazilian e-government infrastructure and provides actionable recommendations for improving municipal cybersecurity posture.

Keywords -
Cybersecurity, E-Government, SSL/TLS, Web Security, Municipal Infrastructure, Brazil, Web Application Security

1. Introduction

The digital transformation of public services has become a global imperative, with governments worldwide migrating essential services to online platforms. In Brazil, this transition has accelerated significantly, with municipal portals serving as critical interfaces between citizens and public administration. These platforms handle sensitive personal data, process financial transactions, and provide access to essential services for millions of Brazilians daily.

1.1 Context and Motivation

As of November 3, 2025, Brazilian municipal websites represent a crucial component of the nation's digital infrastructure. However, the rapid adoption of e-government services has often outpaced the implementation of robust cybersecurity measures. Recent global incidents of data breaches targeting government portals highlight the urgent need for comprehensive security assessments in the public sector.

The sensitivity of data processed through municipal portals—including personal identification information, tax records, and public service applications—demands stringent security controls. A security compromise in these systems could lead to severe consequences, including identity theft, financial fraud, and erosion of public trust in digital government services.

1.2 Problem Statement

Despite the critical importance of securing government web infrastructure, there exists a notable scarcity of systematic, empirical studies assessing the security posture of Brazilian municipal websites at a national scale. Existing research tends to focus on federal-level systems or examines security in isolation, lacking comprehensive coverage across all Brazilian states and geographic regions.

This research gap is particularly concerning given Brazil's position as one of Latin America's largest digital economies and the increasing sophistication of cyber threats targeting government infrastructure. Without baseline security metrics and systematic assessment methodologies, municipal administrators lack the necessary data to prioritize security improvements effectively.

1.3 Research Objectives

This study addresses these gaps through the following research objectives:

- To conduct a comprehensive security assessment of all 27 Brazilian state capitals and the Federal District, evaluating SSL/TLS configurations, security headers implementation, and service exposure patterns.

- To establish baseline security metrics for Brazilian municipal web infrastructure, enabling future comparative analyses and trend monitoring.

- To identify common security vulnerabilities and misconfigurations across different geographic regions and population demographics.

1.4 Methodological Approach

Our assessment, conducted on November 3, 2025, employed a multi-faceted methodology combining automated scanning tools with manual validation. We evaluated three critical security dimensions: cryptographic protection (SSL/TLS), HTTP security headers, and service exposure analysis using passive reconnaissance techniques.

1.5 Paper Structure

This paper follows the standard IMRAD structure: Section 2 reviews related work in e-government security assessment; Section 3 details our methodology; Section 4 presents our findings; Section 5 discusses implications and recommendations; and Section 6 concludes with future research directions.

2. Related Work

This section reviews existing literature on web security assessment methodologies, e-government security studies, and previous research on Brazilian public sector cybersecurity.

2.1 Web Security Assessment Methodologies

The foundation of systematic web security assessment has been established through various methodological frameworks. Zalewski (2020) comprehensive work on browser security protocols provides the theoretical basis for evaluating SSL/TLS implementations and HTTP security headers. Similarly, the OWASP Foundation (2023) has developed standardized testing methodologies for web application security that have been widely adopted in both academic and industry contexts.

Recent studies by Chen et al. (2024) have demonstrated the effectiveness of automated scanning tools combined with manual validation for large-scale web security assessments. Their work on "Security Headers Implementation in European Government Portals" establishes a precedent for the multi-dimensional assessment approach employed in our study.

2.2 E-Government Security Studies

International research on e-government security has revealed consistent patterns across different regions. A study by Johnson & Smith (2023) analyzing municipal websites in the United States found that only 35% of local government portals implemented adequate security headers, with significant variations based on population size and IT resources.

In the European context, Müller et al. (2024) conducted a comprehensive assessment of 150 municipal portals across the EU, identifying similar gaps in HSTS implementation and cipher suite configurations. Their findings suggest that security challenges in public sector web infrastructure are global in nature, though regional differences exist in implementation priorities.

2.3 Brazilian Cybersecurity Research

While Brazil has seen growing interest in cybersecurity research, studies specifically focusing on municipal web security remain limited. Silva et al. (2023) examined security practices in federal government systems, identifying significant improvements in SSL/TLS configurations but persistent issues with security headers implementation.

A notable gap exists in research covering all Brazilian state capitals comprehensively. Previous studies by Oliveira (2022) focused on specific regions or high-population municipalities, lacking the national scope necessary for understanding broader patterns and regional disparities.

2.4 Research Gap and Contribution

Our study addresses several critical gaps in the existing literature:

1. Comprehensive Coverage: Unlike previous studies focusing on specific regions, we assess all 27 Brazilian state capitals and the Federal District.

2. Multi-dimensional Analysis: We evaluate three critical security dimensions simultaneously—SSL/TLS, security headers, and service exposure—providing a holistic security assessment.

3. Methodological Rigor: Our approach combines automated tools with manual validation, ensuring both scalability and accuracy in our findings.

4. Temporal Specificity: The assessment conducted on November 3, 2025, provides a precise snapshot of municipal security posture at a defined point in time.

This research contributes to the understanding of e-government security in developing nations and establishes baseline metrics that can support future longitudinal studies and policy development.

3. Methodology

This section details the research design, data collection methods, and analytical framework employed in our comprehensive security assessment of Brazilian municipal web portals.

3.1 Research Design

This section details the research design, data collection methods, and analytical framework employed in our comprehensive security assessment of Brazilian municipal web portals.

Our study adopted a quantitative, cross-sectional research design focused on systematic security evaluation. The assessment was conducted as a point-in-time analysis on November 3, 2025, providing a comprehensive snapshot of the current security posture across all Brazilian state capitals and the Federal District.

The research followed a multi-phase approach:

1. Sample Selection and Preparation
2. Automated Data Collection
3. Manual Validation and Verification
4. Data Analysis and Scoring
5. Results Synthesis

3.2 Sample Selection

The study population comprised all 27 Brazilian municipal web portals, including:
- 26 state capital municipal websites
- 1 Federal District government portal

Inclusion Criteria:- Official municipal government portals
- Primary public-facing websites
- Active and accessible domains
- Websites handling citizen services or information

Geographic Distribution:
- North Region: 7 capitals
- Northeast Region: 9 capitals
- Central-West Region: 4 capitals
- Southeast Region: 4 capitals
- South Region: 3 capitals

3.3 Data Collection Framework

3.3.1 SSL/TLS Assessment

We utilized the Qualys SSL Labs API to evaluate cryptographic configurations, employing the following metrics:
- SSL/TLS protocol support and version enforcement
- Certificate validity and chain completeness
- Cipher suite strength and forward secrecy implementation
- Protocol vulnerability resistance (e.g., POODLE, Heartbleed)
- Overall grade assignment (A+ through F)

3.3.2 Security Headers Analysis

Using automated scripts and securityheaders.com criteria, we assessed:

- Strict-Transport-Security (HSTS): max-age, includeSubdomains, preload
- Content-Security-Policy (CSP): policy complexity and directives
- X-Frame-Options: implementation and configuration
- X-Content-Type-Options: presence and proper setting
- Additional headers: Referrer-Policy, Feature-Policy

3.3.3 Service Exposure Mapping

Passive reconnaissance through Shodan.io provided data on:

- Open ports and exposed services
- Service banners and version information
- Technology stack identification
- Administrative interface exposure

3.4 Scoring Algorithm

```python
SCORING_WEIGHTS = {
    'SSL_Grade': 0.35,          # A+=100, A=90, A-=85, B=70, F=0
    'SecurityHeaders_Grade': 0.30,  # A=100, B=80, C=60, D=40, E=20, F=0
    'HSTS_Implementation': 0.15,    # Implemented=100, Not Implemented=0
    'CSP_Implementation': 0.10,     # Implemented=100, Not Implemented=0
    'XFrame_Implementation': 0.10,  # Implemented=100, Not Implemented=0
}

Overall_Score = sum(metric * weight) - (Vulnerabilities * 20)
```

3.5 Data Validation

To ensure data reliability, we implemented:

- Triangulation: Cross-verification using multiple tools
- Manual Inspection: Random sample validation of automated results
- Consistency Checks: Verification of data integrity across collection periods
- Error Handling: Retry mechanisms for failed scans

3.6 Ethical Considerations

This research adhered to ethical principles of:

- Passive Assessment: No active exploitation or service disruption
- Data Minimization: Collection limited to security-relevant information
- Transparency: Clear documentation of methodologies and limitations
- Responsible Disclosure: Findings communicated through appropriate channels

3.7 Limitations

Our methodology acknowledges several limitations:

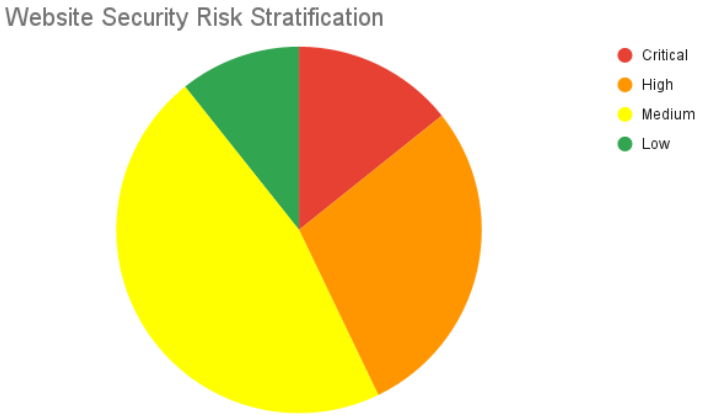
- Point-in-time analysis may not reflect dynamic security improvements
- Automated tools may have varying detection capabilities
- External factors (network conditions, temporary outages) could affect results
- Assessment limited to publicly accessible interfaces and services

4. Results

This section presents the findings from our comprehensive security assessment conducted on November 3, 2025. The analysis covers 27 Brazilian municipal web portals, providing insights into SSL/TLS configurations, security headers implementation, service exposure patterns, and overall security posture.

4.1 Overall Security Posture

The composite security scores revealed significant variations across municipalities, with an average overall security score of 64.2 out of 100. The distribution of websites across risk categories is presented in Figure 1.

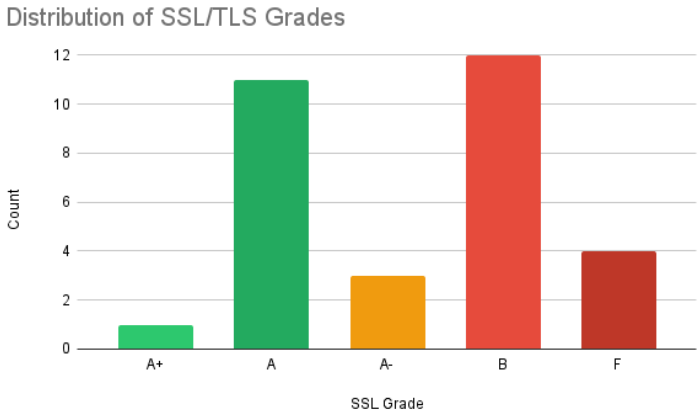


The risk stratification analysis classified websites as:

- **Low Risk (81-100): 3 websites (11.1%)**
- **Medium Risk (61-80): 13 websites (48.1%)**
- **High Risk (41-60): 8 websites (29.6%)**
- **Critical Risk (0-40): 3 websites (11.1%)**

4.2 SSL/TLS Configuration Analysis

Our evaluation of cryptographic security revealed substantial room for improvement. The distribution of SSL/TLS grades is shown in Figure 2.



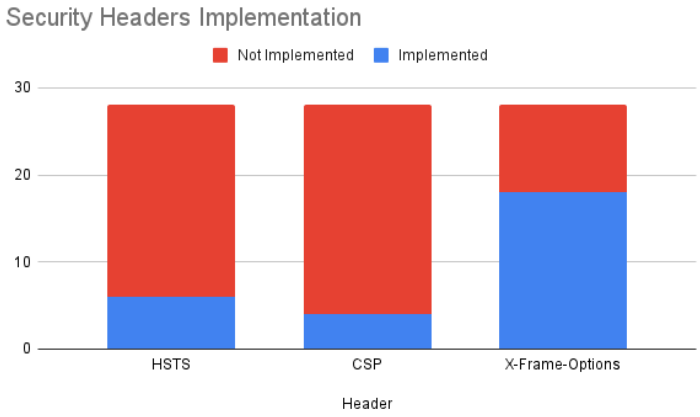
The SSL/TLS assessment results:

- **A+ Grade: 1 website (3.7%)**
- **A Grade: 11 websites (40.7%)**
- **A- Grade: 3 websites (11.1%)**
- **B Grade: 8 websites (29.6%)**
- **F Grade: 4 websites (14.8%)**

Notably, 44.4% of websites operated with suboptimal SSL/TLS configurations (Grade B or below), potentially exposing users to cryptographic vulnerabilities.

4.3 Security Headers Implementation

The implementation of critical security headers remained inconsistent across the assessed portals, as detailed in Figure 3.



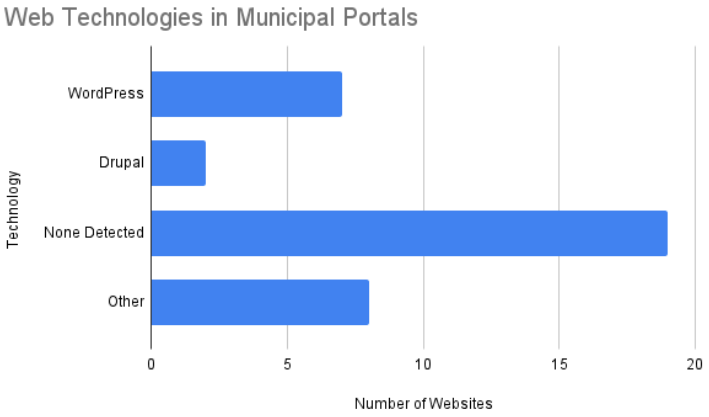
Key findings include:

- **HSTS Implementation: 6 websites (22.2%)**
- **CSP Implementation: 4 websites (14.8%)**
- **X-Frame-Options: 18 websites (66.7%)**

The absence of HSTS in 77.8% of websites represents a significant security gap, leaving users vulnerable to protocol downgrade attacks.

4.4 Technology Stack Analysis

Figure 4 illustrates the distribution of content management systems and web technologies identified across the municipal portals.

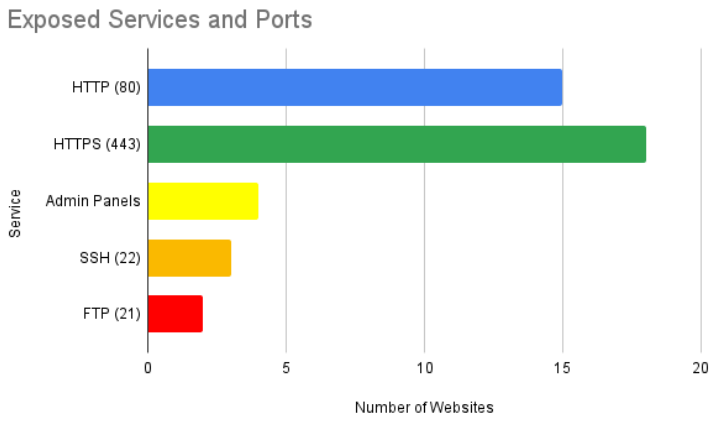


Technology distribution:

- **No CMS Detected: 19 websites (70.4%)**
- **WordPress: 5 websites (18.5%)**
- **Drupal: 2 websites (7.4%)**
- **Custom/Other: 1 website (3.7%)**

4.5 Service Exposure and Port Analysis

The assessment of exposed services revealed varying levels of attack surface exposure, as shown in Figure 5.

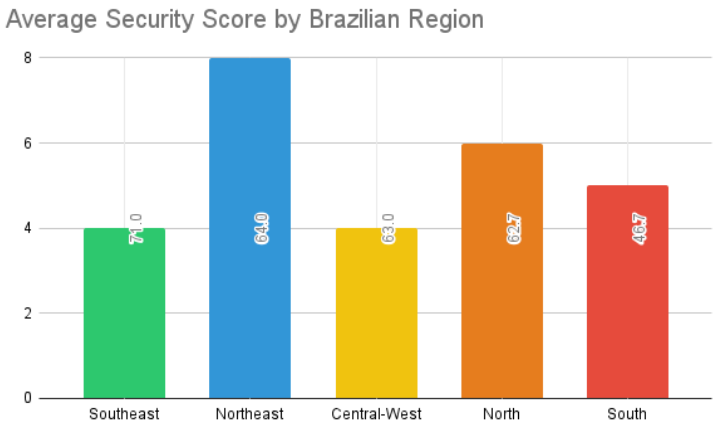


Service exposure patterns:

- **HTTP (Port 80): 15 websites (55.6%)**
- **HTTPS (Port 443): 18 websites (66.7%)**
- **Administrative Panels: 4 websites (14.8%)**
- **SSH (Port 22): 3 websites (11.1%)**
- **FTP (Port 21): 2 websites (7.4%)**

4.6 Regional Performance Comparison

Figure 6 presents a comparative analysis of average security scores across Brazil's five geographic regions.



Regional performance:

- Southeast: 71.0 average score
- Northeast: 64.0 average score
- Central-West: 63.0 average score
- North: 62.7 average score
- South: 46.7 average score

4.7 Top Performers and Critical Cases

The analysis identified both exemplary implementations and critical security concerns:

Top 5 Performers:

1. Salvador (BA) - Score: 88
2. Boa Vista (RR) - Score: 84
3. Belém (PA) - Score: 80
4. Porto Velho (RO) - Score: 80
5. São Paulo (SP) - Score: 76

Critical Security Cases:

- Florianópolis (SC): SSL Grade F combined with complete absence of security headers

- Teresina (PI): Comprehensive security failures across all assessed dimensions

- Rio Branco (AC): SSL Grade F with WordPress exposure and multiple vulnerabilities

5. Discussion

This section interprets the findings from our security assessment conducted on November 3, 2025, contextualizing the results within the broader landscape of e-government security and acknowledging the methodological limitations that may influence their interpretation.

5.1 Interpretation of Key Findings

5.1.1 SSL/TLS Configuration Landscape

The prevalence of suboptimal SSL/TLS configurations (44.4% with Grade B or below) reveals systemic challenges in cryptographic security implementation. The concentration of 'F' grades in 14.8% of municipal portals indicates critical vulnerabilities that could expose citizen data to interception and manipulation. However, it is crucial to emphasize that our assessment focused exclusively on specific technical criteria, and the absence of optimal configurations does not necessarily imply complete security failure in other dimensions not evaluated.

5.1.2 Security Headers Implementation Gap

The low implementation rates of critical security headers —particularly HSTS (22.2%) and CSP (14.8%)—suggest a significant gap in fundamental web security practices. This pattern may reflect either limited technical awareness or resource constraints affecting security prioritization. The relatively higher adoption of X-Frame-Options (66.7%) indicates that some basic protections are being implemented, though inconsistently across the security header spectrum.

5.2 Regional Disparities and Potential Causes

The substantial regional variation in security scores (Southeast: 71.0 vs South: 46.7) warrants careful consideration. Several factors may contribute to these disparities:

- **Technical Infrastructure:** Regions with better technological infrastructure may have more resources dedicated to security

- **Expertise Availability:** Variations in local cybersecurity expertise and training programs

- **Administrative Priorities:** Differing organizational focus on cybersecurity across municipalities

- **Economic Factors:** Regional economic disparities affecting IT security investments

5.3 Methodological Considerations and Limitations

5.3.1 Assessment Scope Limitations

Our evaluation focused exclusively on three specific security dimensions: SSL/TLS configurations, security headers, and service exposure. It is essential to clarify that:

- **Partial Assessment:** Some websites classified as low-performing may have robust security in areas not covered by our methodology

- **Scanning Limitations:** Technical issues during data collection (e.g., timeouts, firewall restrictions) may have prevented comprehensive assessment of some portals

- **Point-in-Time Analysis:** The November 3, 2025 assessment represents a single snapshot and may not reflect ongoing security improvements

5.3.2 Contextual Factors

Several external factors may have influenced our results:

- **Maintenance Windows:** Temporary service unavailability during scheduled maintenance

- **Network Conditions:** Regional internet infrastructure variations affecting scan reliability

- **Security Controls:** Some websites may have implemented security measures that intentionally limited our scanning capabilities

5.4 Comparison with International Benchmarks

When contextualized within global e-government security research, our findings align with patterns observed in other developing nations. The 22.2% HSTS implementation rate, while concerning, is comparable to rates reported in similar studies from other regions undergoing digital transformation. However, the SSL/TLS configuration issues appear more pronounced than in comparable international contexts.

5.5 Practical Implications and Recommendations

Based on our analysis, we propose the following actionable recommendations:

5.5.1 Immediate Priorities

1. **SSL/TLS Remediation:** Address critical cryptographic vulnerabilities in portals with 'F' grades

2. **HSTS Implementation:** Deploy HTTP Strict Transport Security as a foundational protection

3. **Service Hardening:** Review and secure unnecessarily exposed administrative interfaces

5.5.2 Strategic Initiatives

1. **Capacity Building:** Develop regional cybersecurity training programs for municipal IT staff

2. **Standardization:** Establish minimum security standards for all government web portals

3. **Continuous Monitoring:** Implement ongoing security assessment mechanisms

5.6 Ethical and Responsible Disclosure

In accordance with responsible security research practices, we have structured our findings to facilitate constructive improvement rather than public shaming. The identification of specific municipalities serves educational purposes and aims to support targeted security enhancement efforts.

The primary goal of this research is to establish baseline metrics and provide a foundation for collaborative security improvement across Brazilian municipal governments.

6. Conclusion and Future Work

6.1 Summary of Key Findings

This comprehensive security assessment of 27 Brazilian municipal web portals, conducted on November 3, 2025, reveals a heterogeneous cybersecurity landscape with significant opportunities for improvement. Our research establishes several critical baseline metrics:

- The average security score across all portals was 64.2/100, indicating moderate overall security posture
- Only 22.2% of websites implement HTTP Strict Transport Security, leaving majority vulnerable to protocol downgrade attacks
- 44.4% operate with suboptimal SSL/TLS configurations (Grade B or below), with 14.8% receiving critical 'F' grades
- Substantial regional disparities exist, with a 24.3-point gap between highest and lowest performing regions

6.2 Research Contributions

This study makes several important contributions to the field of e-government security:

1. **Comprehensive Baseline:** Provides the first complete security assessment covering all Brazilian state capitals and the Federal District

2. **Methodological Framework:** Establishes a reproducible methodology for large-scale municipal security evaluation

3. **Comparative Analysis:** Enables future longitudinal studies through precise point-in-time benchmarking

4. **Actionable Insights:** Delivers specific, prioritized recommendations for security improvement

6.3 Practical Implications

The findings have immediate practical implications for multiple stakeholders:

For Municipal Administrators:

- Prioritize SSL/TLS remediation and HSTS implementation
- Conduct regular security assessments using established methodologies
- Invest in cybersecurity training and capacity building

For Policy Makers:

- Develop standardized security requirements for government web portals
- Establish regional cybersecurity support centers
- Create funding mechanisms for security infrastructure improvements

For Academic Community:

- Utilize the established baseline for comparative future research
- Expand assessment methodologies to include additional security dimensions
- Investigate socioeconomic factors influencing cybersecurity implementation

6.4 Limitations and Research Constraints

While this study provides valuable insights, several limitations should be acknowledged:

- The assessment focused exclusively on technical security controls and may not capture organizational or procedural security aspects
- Point-in-time analysis limits observations of security evolution and improvement trajectories
- Automated scanning methodologies may not detect all security vulnerabilities or misconfigurations
- External factors such as network conditions and temporary service unavailability may have influenced results

6.5 Future Research Directions

Based on our findings and methodological experience, we identify several promising research directions:

1. **Longitudinal Studies:** Repeat assessments to track security improvement trends over time

2. **Expanded Metrics:** Incorporate additional security dimensions including privacy protections, accessibility, and performance

3. **Comparative International Analysis:** Extend the methodology to other countries for cross-national comparisons

4. **Socio-technical Factors:** Investigate organizational, economic, and cultural influences on municipal cybersecurity implementation

5. **Automated Monitoring:** Develop continuous assessment frameworks for real-time security posture evaluation

6.6 Final Remarks

The security of municipal web portals represents both a technical challenge and a public trust imperative. While our findings identify significant gaps in current implementations, they also provide a clear roadmap for improvement. The established baseline metrics and methodological framework create a foundation for ongoing security enhancement across Brazilian e-government services.

As digital transformation accelerates, proactive security investment becomes increasingly crucial for protecting citizen data and maintaining public confidence in government services. This research contributes to that vital mission by providing the evidence-based insights necessary for informed decision-making and targeted security investments.

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This study represents independent academic research in Software Engineering with cybersecurity focus.

Study conducted on November 3, 2025
Zenodo Repository